

	MIDTERM EXAMINATION FALL 2007 CS504 - SOFTWARE ENGINEERING – I (Session - 7)	Marks: 40 Time: 120min
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StudentID/LoginID: _____

Student Name: _____

Center Name/Code: _____

Exam Date: Thursday, November 22, 2007

Please read the following instructions carefully before attempting any of the questions:

1. Attempt all questions. Marks are written adjacent to each question.
2. Do not ask any questions about the contents of this examination from anyone.
 - a. If you think that there is something wrong with any of the questions, attempt it to the best of your understanding.
 - b. If you believe that some essential piece of information is missing, make an appropriate assumption and use it to solve the problem.
 - c. Write all steps, missing steps may lead to deduction of marks.
3. The use of Mobile phone is forbidden during the paper.

****WARNING: Please note that Virtual University takes serious note of unfair means.**

Anyone found involved in cheating will get an 'F' grade in this course

For Teacher's use only										
Question Marks	1	2	3	4	5	6	7			Total

Question No: 1 (Marks: 2) - Please choose one

Business model is the part of designing phase of software engineering development life cycle

► True

▶ False

Question No: 2 (Marks: 2) - Please choose one

The state transition diagram

- ▶ depicts relationships between data objects
- ▶ depicts functions that transform the data flow
- ▶ indicates how data are transformed by the system
- ▶ indicates system reactions to external events

Question No: 3 (Marks: 2) - Please choose one

The best way to conduct a requirements validation review is to

- ▶ Examine the system
- ▶ Have the customer look over the requirements
- ▶ Send them to the design team
- ▶ Use checklist to examine the questions

Question No: 4 (Marks: 2) - Please choose one

The system specification describes the

- ▶ Function, performance and constraints of a computer-based system
- ▶ implementation of each allocated system

- ▶ time required for simulation
- ▶ element software architecture

Question No: 5 (Marks: 12)

List three necessary and one desired property of requirements: [4 Marks]

For each of the following given Requirement indicate whether it is Functional (F) or Non-Functional (N) Requirement. [2 marks each]

- 1. The System should be designed so it can later be extended to a Blue Line reservation System.**
- 2. The System will allow the user to view the layout of seats in Bus.**
- 3. The System must be available at all times.**
- 4. A given seat can be assigned to only one person.**

Question No: 6 (Marks: 10)

Requirement engineering mainly deals with the definition phase of the system. Discuss the importance of Requirement Engineering.

Question No: 7 (Marks: 10)

Identify appropriate classes and methods for the following requirements specification, and draw the resulting class diagram:

“Customers order a number of items from a shop. The purpose of this system is to produce a printed invoice for the customers’ orders. Each invoice should contain a full list of items ordered together with their price. Customers might order more than one of a particular item so the invoice should show the number of each item ordered rather than list them multiple times. The invoice should display the cost of each item and the total amount payable. The invoice should also print the customer’s name and address.”